Utah wildfire occurrence modeling

Adam Valentino | ML v1 publication 1.0.0 | 16 October 2026

Development 2001-2021. Held-out annual maps 2013-2021. The target is at least one recorded wildfire discovery within a county-clipped cell during a UTC year. The 997 nominal 4 km cells cover Salt Lake, Utah and Washington counties, including non-WUI land.

Development evidence, not independent final evaluation. These years informed model and calibration choices. Individual observations were held out from their fitted models, but the reported comparison is retrospective model development. No claim of statistical significance, statewide transfer or operational readiness is made.

Frozen model decision

Conservative XGBoost is the probability recommendation. Raw XGBoost is the development ranking reference; sigmoid Random Forest is the benchmark. The maps combine five geographic fits per year under one frozen specification.

Mean annual Brier score: **0.070814** conservative XGBoost, **0.071353** raw XGBoost and **0.071341** sigmoid Random Forest. The relative reduction in mean Brier score versus past-rate references is about 1%.

Raw top-decile recall averages **18.09%** per year, versus **17.42%** for conservative probabilities. Probability calibration and location ranking have different trade-offs.

Spatial and temporal safeguards

For target year T, base training uses 2001 through T-3; calibration uses T-2 and T-1. Five spatial folds use 20 km blocks with 4 km exclusion buffers. Imputation is fitted on training rows only. There are 8,588 eligible held-out cell-years and 656 recorded positives.

Observation area matters

Full 16 km2 cells account for **75.0%** of eligible cell-years and **98.0%** of top-decile slots. The clipped-cell conservative mean is **3.22%**, versus **3.03%** recorded occurrence. The larger clipped group underpredicts: **6.40% versus 8.09%**.

Changing only the area input to 16 km2 raises the clipped-cell mean to **7.61%** and replaces **96/863** conservative selections. This is a sensitivity diagnostic, not a corrected forecast. **All maps retain original values and rankings.**

Remaining limitations

Full cells account for **89.2%** of the 2020 count shortfall. Reporting changes and real occurrence changes cannot be separated. No-record labels do not establish absence of fire.

Contemporary terrain is a static historical proxy. Climate uses pre-cutoff windows and an assumed 60-day lag. Historical as-issued terrain and climate availability is unverified. Probabilities are not property-loss estimates, daily fire danger or per-km2 hazard.

Frozen baseline and model unchanged. No new training or 2024-2025 evaluation. Text, tables and map descriptions are also available in the accompanying offline index.html.

Sources and map reading

Source metadata checked 16 October 2026 | Analysis uses retained snapshots

Recorded wildfire labels

NIFC InFORM public fire-occurrence layer; reviewed development labels for 2001-2021. CreatedBySystem and reporting-status changes around 2020 are descriptive evidence, not proof of a change in true occurrence or completeness. Raw incident records are not redistributed.
https://www.nifc.gov/programs/information-technology

Terrain

USGS 3DEP / National Elevation Dataset 1 arc-second catalog retained 15 October 2026; aligned 30 m terrain. Catalog publication dates span 2013-2026 and are not terrain observation dates. Historical as-issued availability is not established.
https://www.usgs.gov/3d-elevation-program

County footprint

State of Utah, SGID; UGRC and Lieutenant Governor's Office. Retained county response acquired 15 October 2026. Administrative boundaries define the fixed study footprint, not cadastral accuracy.
https://gis.utah.gov/products/sgid/boundaries/county/

Antecedent climate

John T. Abatzoglou, gridMET (2013), International Journal of Climatology 33, 121-131, doi:10.1002/joc.3413. Subset archive completed 16 October 2026. Source days 1 October 1999 through 30 September 2020 support the 2001-2021 targets.
https://www.climatologylab.org/gridmet.html

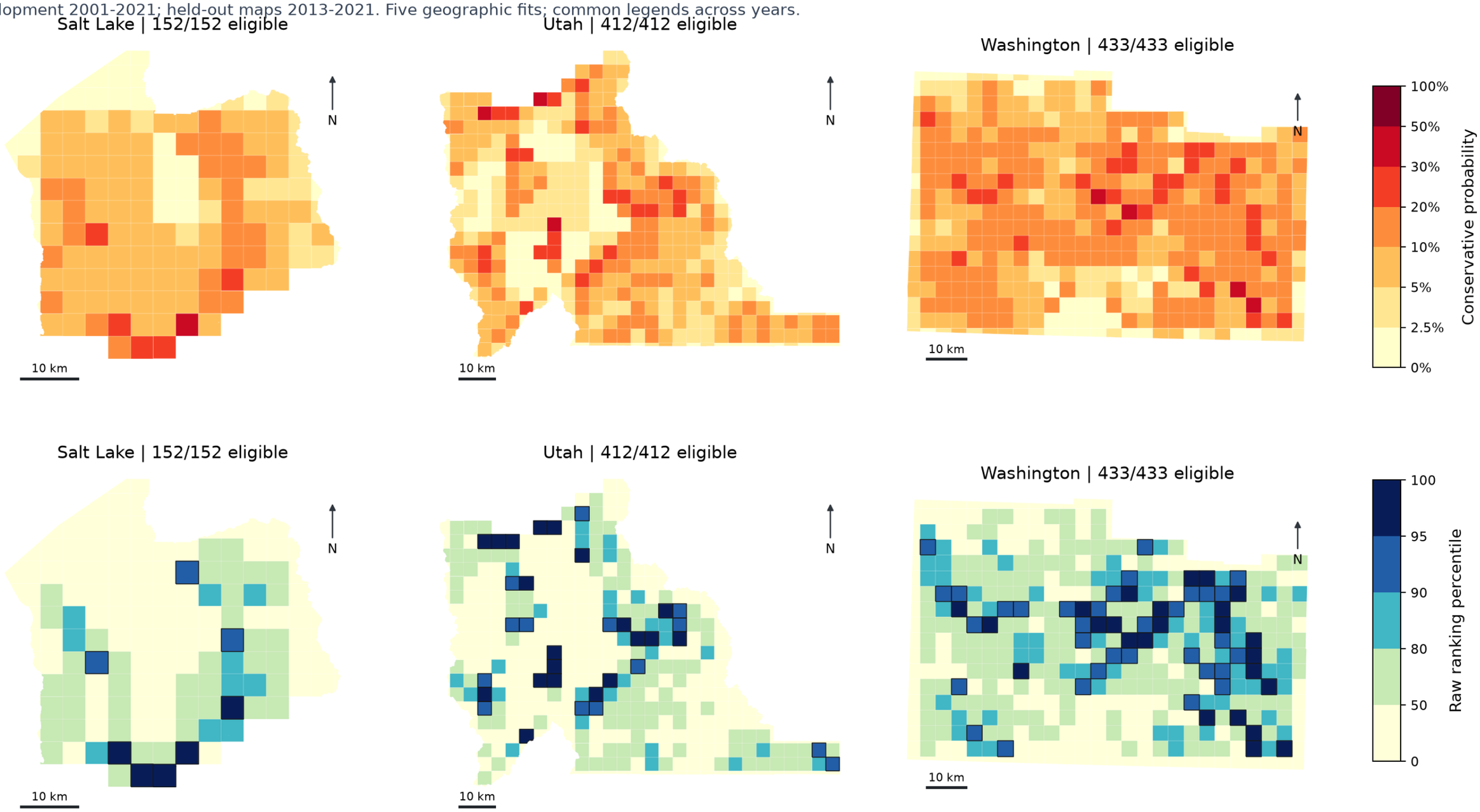
Read the maps

Top panels: conservative recorded-occurrence probability with common legends. Bottom panels: raw ranking percentile, 100 highest. Outlines select ceil(10% of eligible cells) across all three counties each year. Hatching identifies unknown labels or excluded flagged zeros, not low risk. County panels have separate extents; coordinates use EPSG:26912.

Every following page includes the unchanged map image and a searchable text description. County averages describe eligible cells only. Native text companions improve access; this PDF is not represented as a fully tagged PDF/UA document. The offline HTML provides semantic headings, descriptions and tables.

2013 | Historical held-out wildfire occurrence

Top: conservative annual recorded-occurrence probability per cell. Bottom: raw ranking percentile, 100 = highest.
Development 2001-2021; held-out maps 2013-2021. Five geographic fits; common legends across years.



Eligible: 997/997. Excluded: 0 unknown labels, 0 flagged zeros. Raw top 10%: 100 cells across all three counties.
Area audit, pooled 2013-2021: full 16 km2 cells = 75.0% of eligible cell-years and 98.0% of top-10% selections.
Cell probabilities depend on clipped area; they are not per-km2 hazard. Original probabilities and ranking inputs are unchanged.
Historical development; county footprint includes non-WUI land. Reporting and climate-vintage limits apply. Separate panel extents; EPSG:26912.

2013 accessible map description

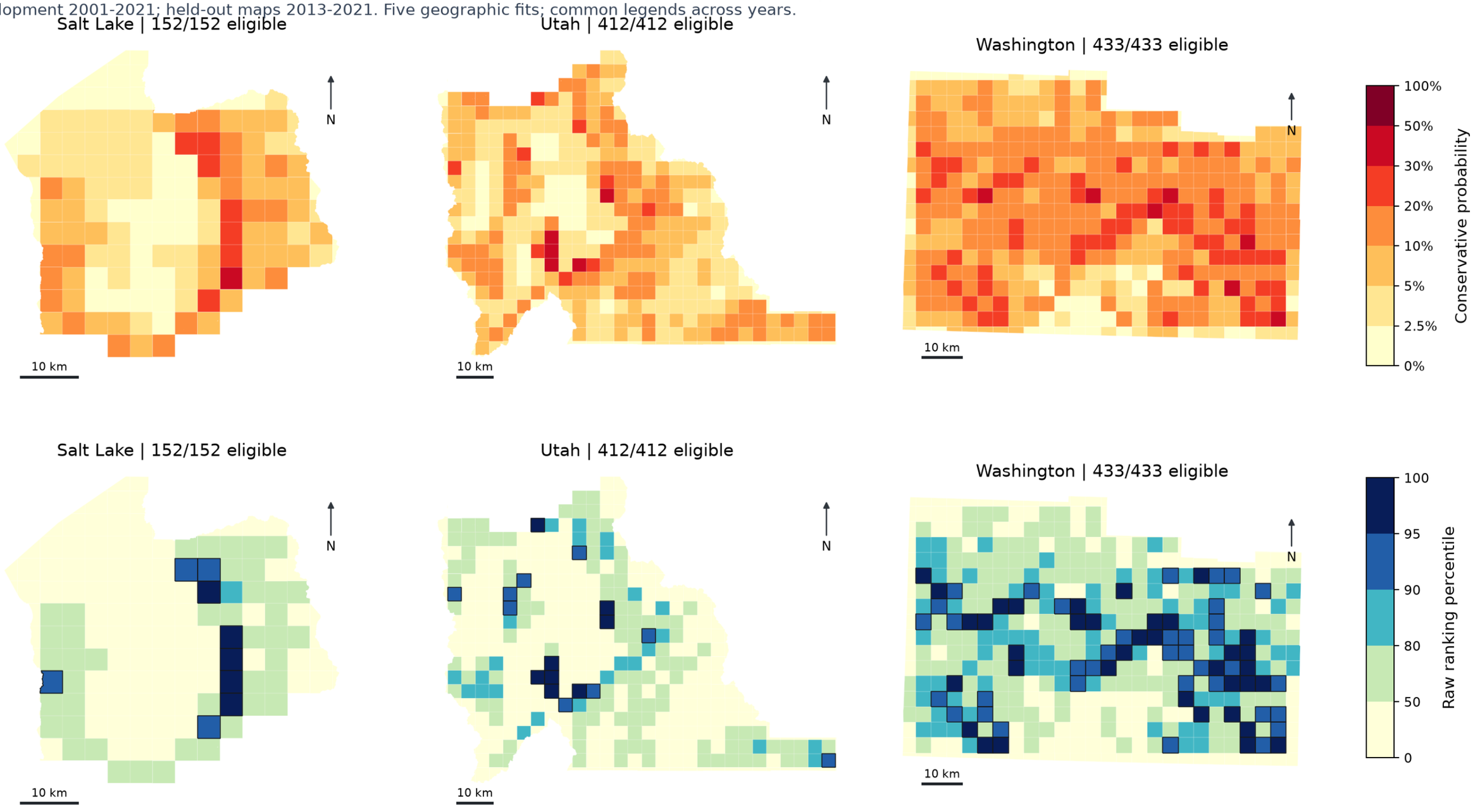
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2013: 997 of 997 original cells have held-out predictions; 0 unknown-label and 0 flagged-zero cells are excluded. Outlines select 100 cells across the three-county cohort. In this year, full cells represent 75.6% of eligible cells and 99.0% of raw top-decile selections. Upper panels show conservative probabilities; lower panels show raw ranking percentiles. Original-area values are retained, and boundary-cell probabilities are not per-km² hazard measures. Development years informed model selection; terrain is a contemporary static proxy.

Geographic comparison among eligible cells: Salt Lake has mean conservative probability 7.19% and 8 raw top-decile cells; Utah has mean conservative probability 7.59% and 34 raw top-decile cells; Washington has mean conservative probability 10.59% and 58 raw top-decile cells. County averages do not imply uniform conditions within a county; the mapped pattern is clustered, and panel scales differ.

2014 | Historical held-out wildfire occurrence

Top: conservative annual recorded-occurrence probability per cell. Bottom: raw ranking percentile, 100 = highest.
Development 2001-2021; held-out maps 2013-2021. Five geographic fits; common legends across years.



Eligible: 997/997. Excluded: 0 unknown labels, 0 flagged zeros. Raw top 10%: 100 cells across all three counties.
Area audit, pooled 2013-2021: full 16 km2 cells = 75.0% of eligible cell-years and 98.0% of top-10% selections.
Cell probabilities depend on clipped area; they are not per-km2 hazard. Original probabilities and ranking inputs are unchanged.
Historical development; county footprint includes non-WUI land. Reporting and climate-vintage limits apply. Separate panel extents; EPSG:26912.

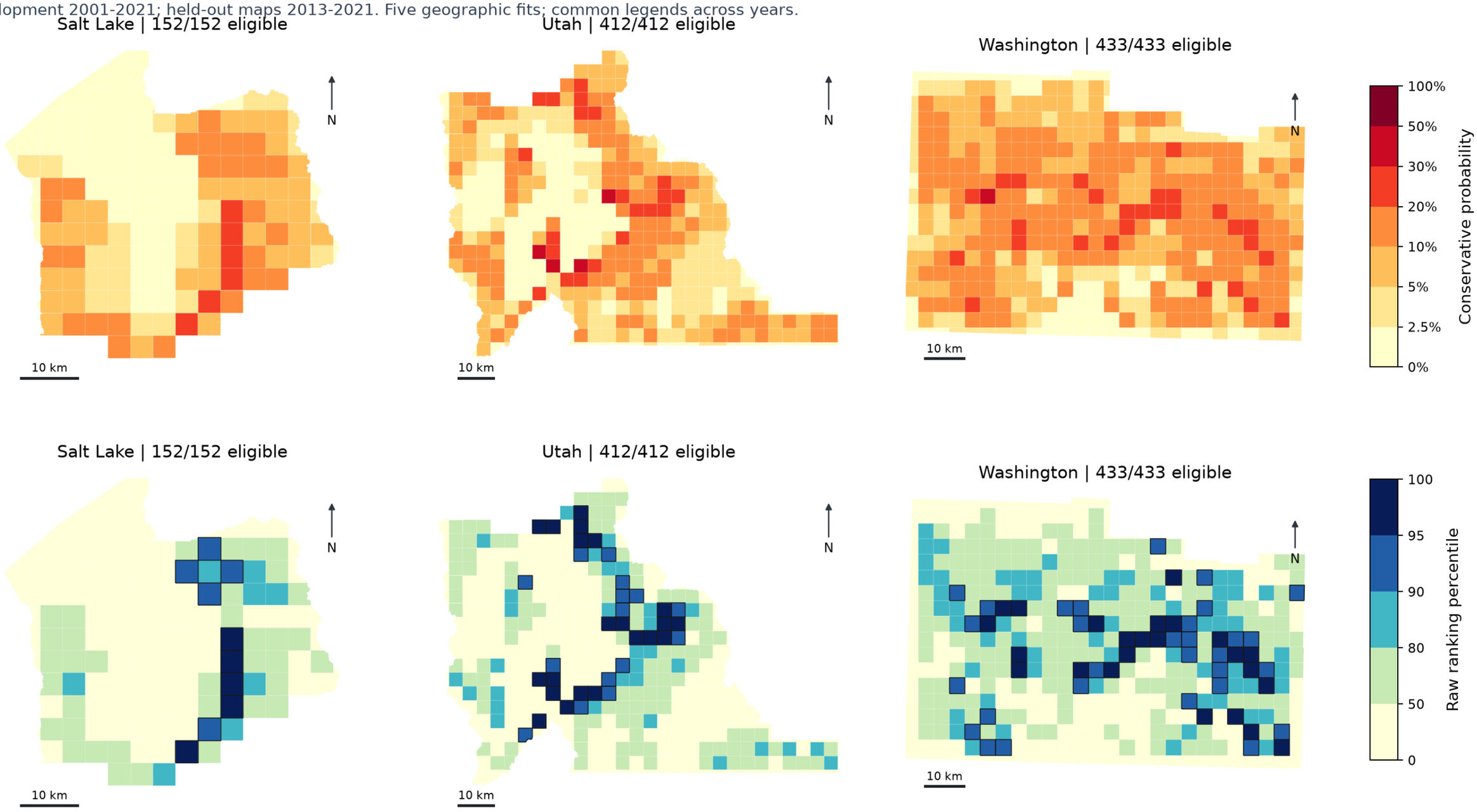
2014 accessible map description

2014: 997 of 997 original cells have held-out predictions; 0 unknown-label and 0 flagged-zero cells are excluded. Outlines select 100 cells across the three-county cohort. In this year, full cells represent 75.6% of eligible cells and 99.0% of raw top-decile selections. Upper panels show conservative probabilities; lower panels show raw ranking percentiles. Original-area values are retained, and boundary-cell probabilities are not per-km² hazard measures. Development years informed model selection; terrain is a contemporary static proxy.

Geographic comparison among eligible cells: Salt Lake has mean conservative probability 5.99% and 9 raw top-decile cells; Utah has mean conservative probability 6.93% and 17 raw top-decile cells; Washington has mean conservative probability 12.01% and 74 raw top-decile cells. County averages do not imply uniform conditions within a county; the mapped pattern is clustered, and panel scales differ.

2015 | Historical held-out wildfire occurrence

Top: conservative annual recorded-occurrence probability per cell. Bottom: raw ranking percentile, 100 = highest.
Development 2001-2021; held-out maps 2013-2021. Five geographic fits; common legends across years.



Eligible: 997/997. Excluded: 0 unknown labels, 0 flagged zeros. Raw top 10%: 100 cells across all three counties.
Area audit, pooled 2013-2021: full 16 km2 cells = 75.0% of eligible cell-years and 98.0% of top-10% selections.
Cell probabilities depend on clipped area; they are not per-km2 hazard. Original probabilities and ranking inputs are unchanged.
Historical development; county footprint includes non-WUI land. Reporting and climate-vintage limits apply. Separate panel extents; EPSG:26912.

2015 accessible map description

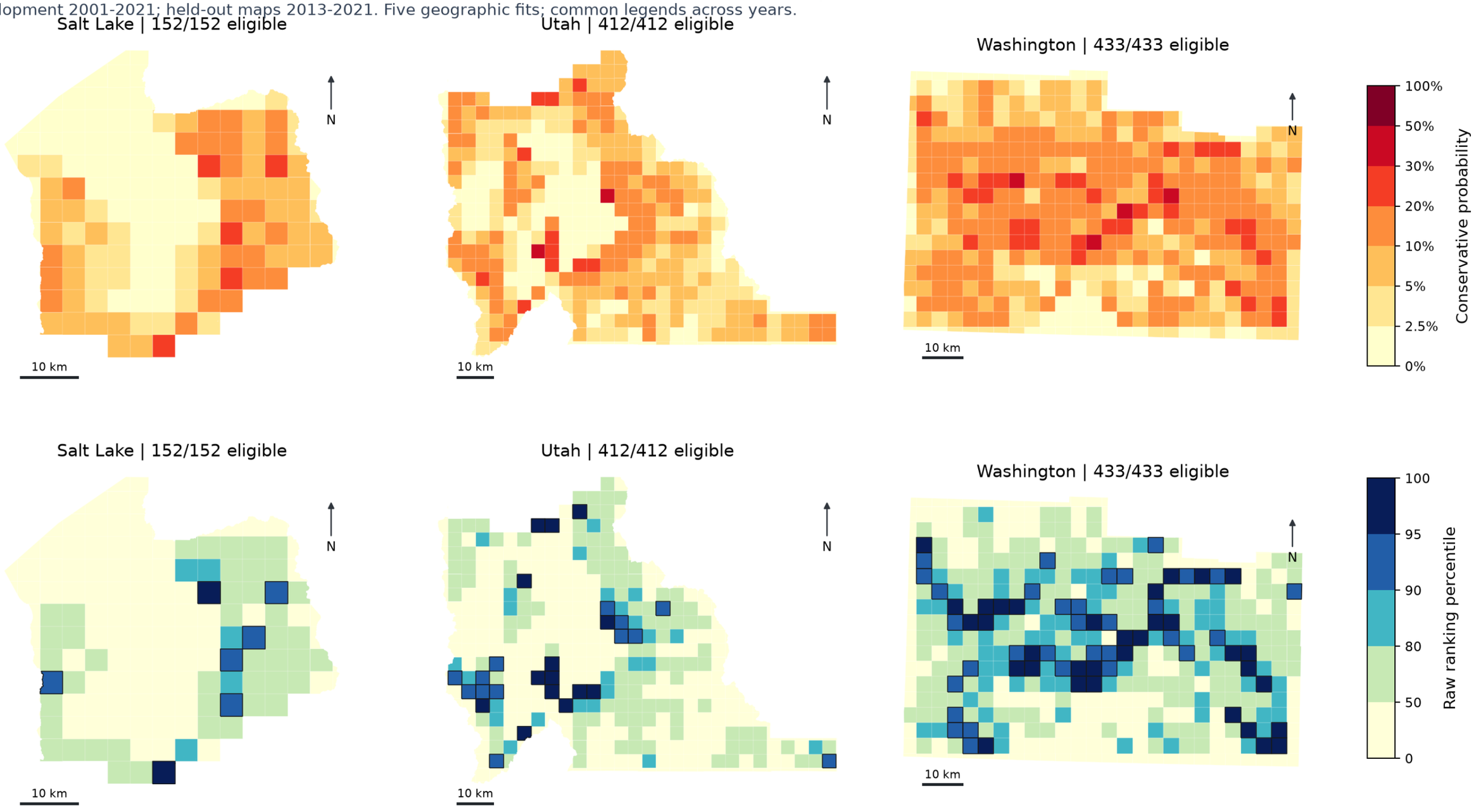
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2015: 997 of 997 original cells have held-out predictions; 0 unknown-label and 0 flagged-zero cells are excluded. Outlines select 100 cells across the three-county cohort. In this year, full cells represent 75.6% of eligible cells and 98.0% of raw top-decile selections. Upper panels show conservative probabilities; lower panels show raw ranking percentiles. Original-area values are retained, and boundary-cell probabilities are not per-km² hazard measures. Development years informed model selection; terrain is a contemporary static proxy.

Geographic comparison among eligible cells: Salt Lake has mean conservative probability 6.03% and 10 raw top-decile cells; Utah has mean conservative probability 7.26% and 36 raw top-decile cells; Washington has mean conservative probability 9.58% and 54 raw top-decile cells. County averages do not imply uniform conditions within a county; the mapped pattern is clustered, and panel scales differ.

2016 | Historical held-out wildfire occurrence

Top: conservative annual recorded-occurrence probability per cell. Bottom: raw ranking percentile, 100 = highest.
Development 2001-2021; held-out maps 2013-2021. Five geographic fits; common legends across years.



Eligible: 997/997. Excluded: 0 unknown labels, 0 flagged zeros. Raw top 10%: 100 cells across all three counties.
Area audit, pooled 2013-2021: full 16 km2 cells = 75.0% of eligible cell-years and 98.0% of top-10% selections.
Cell probabilities depend on clipped area; they are not per-km2 hazard. Original probabilities and ranking inputs are unchanged.
Historical development; county footprint includes non-WUI land. Reporting and climate-vintage limits apply. Separate panel extents; EPSG:26912.

2016 accessible map description

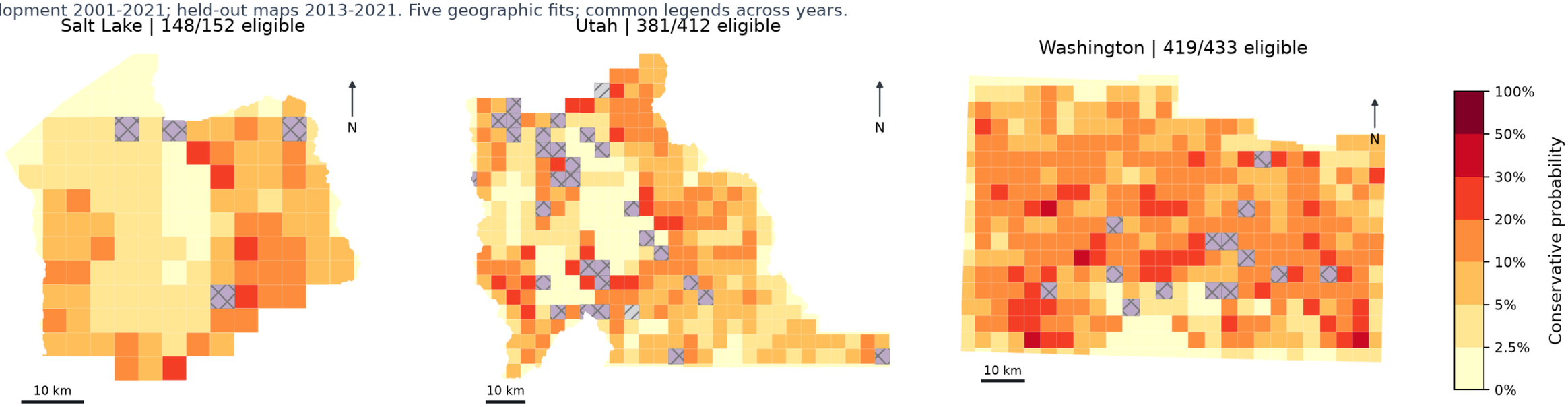
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2016: 997 of 997 original cells have held-out predictions; 0 unknown-label and 0 flagged-zero cells are excluded. Outlines select 100 cells across the three-county cohort. In this year, full cells represent 75.6% of eligible cells and 96.0% of raw top-decile selections. Upper panels show conservative probabilities; lower panels show raw ranking percentiles. Original-area values are retained, and boundary-cell probabilities are not per-km² hazard measures. Development years informed model selection; terrain is a contemporary static proxy.

Geographic comparison among eligible cells: Salt Lake has mean conservative probability 5.61% and 7 raw top-decile cells; Utah has mean conservative probability 6.32% and 27 raw top-decile cells; Washington has mean conservative probability 9.95% and 66 raw top-decile cells. County averages do not imply uniform conditions within a county; the mapped pattern is clustered, and panel scales differ.

2017 | Historical held-out wildfire occurrence

Top: conservative annual recorded-occurrence probability per cell. Bottom: raw ranking percentile, 100 = highest.
Development 2001-2021; held-out maps 2013-2021. Five geographic fits; common legends across years.



Eligible: 948/997. Excluded: 2 unknown labels, 47 flagged zeros. Raw top 10%: 95 cells across all three counties.
Area audit, pooled 2013-2021: full 16 km2 cells = 75.0% of eligible cell-years and 98.0% of top-10% selections.
Cell probabilities depend on clipped area; they are not per-km2 hazard. Original probabilities and ranking inputs are unchanged.
Historical development; county footprint includes non-WUI land. Reporting and climate-vintage limits apply. Separate panel extents; EPSG:26912.

2017 accessible map description

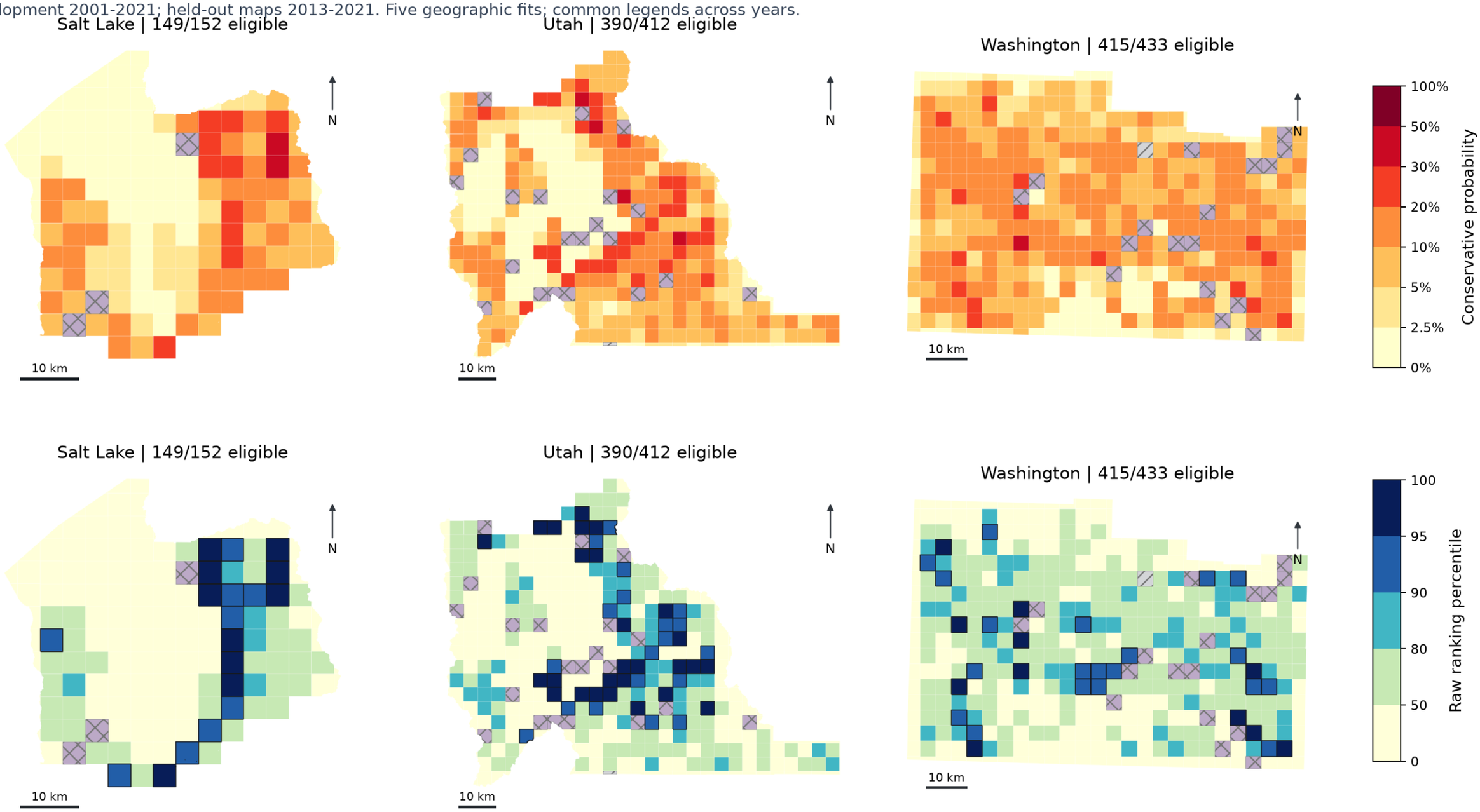
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2017: 948 of 997 original cells have held-out predictions; 2 unknown-label and 47 flagged-zero cells are excluded. Outlines select 95 cells across the three-county cohort. In this year, full cells represent 74.8% of eligible cells and 97.9% of raw top-decile selections. Upper panels show conservative probabilities; lower panels show raw ranking percentiles. Original-area values are retained, and boundary-cell probabilities are not per-km² hazard measures. Development years informed model selection; terrain is a contemporary static proxy.

Geographic comparison among eligible cells: Salt Lake has mean conservative probability 6.07% and 9 raw top-decile cells; Utah has mean conservative probability 6.63% and 24 raw top-decile cells; Washington has mean conservative probability 10.66% and 62 raw top-decile cells. County averages do not imply uniform conditions within a county; the mapped pattern is clustered, and panel scales differ.

2018 | Historical held-out wildfire occurrence

Top: conservative annual recorded-occurrence probability per cell. Bottom: raw ranking percentile, 100 = highest.
Development 2001-2021; held-out maps 2013-2021. Five geographic fits; common legends across years.



Eligible: 954/997. Excluded: 2 unknown labels, 41 flagged zeros. Raw top 10%: 96 cells across all three counties.
Area audit, pooled 2013-2021: full 16 km2 cells = 75.0% of eligible cell-years and 98.0% of top-10% selections.
Cell probabilities depend on clipped area; they are not per-km2 hazard. Original probabilities and ranking inputs are unchanged.
Historical development; county footprint includes non-WUI land. Reporting and climate-vintage limits apply. Separate panel extents; EPSG:26912.

2018 accessible map description

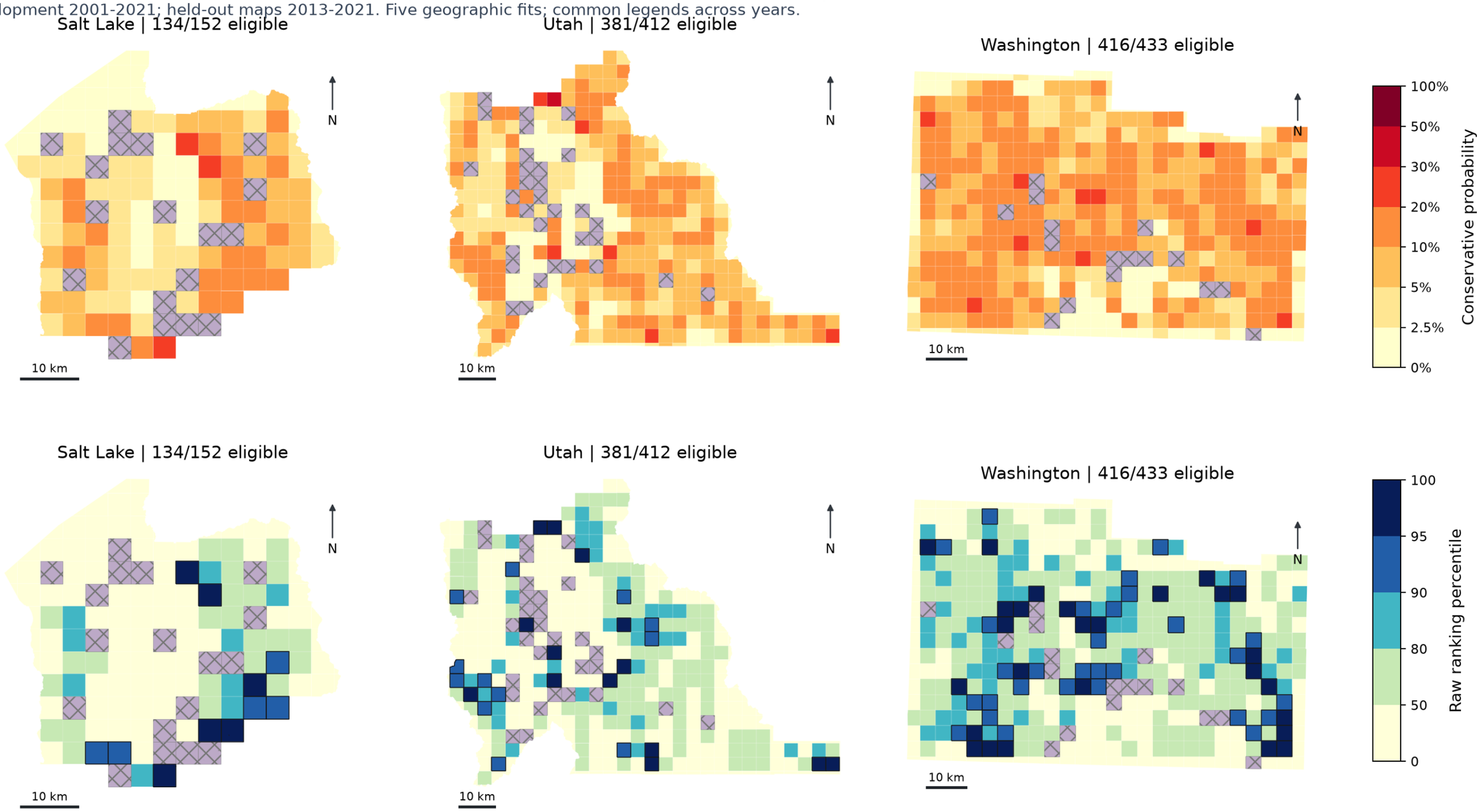
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2018: 954 of 997 original cells have held-out predictions; 2 unknown-label and 41 flagged-zero cells are excluded. Outlines select 96 cells across the three-county cohort. In this year, full cells represent 75.3% of eligible cells and 97.9% of raw top-decile selections. Upper panels show conservative probabilities; lower panels show raw ranking percentiles. Original-area values are retained, and boundary-cell probabilities are not per-km² hazard measures. Development years informed model selection; terrain is a contemporary static proxy.

Geographic comparison among eligible cells: Salt Lake has mean conservative probability 7.14% and 19 raw top-decile cells; Utah has mean conservative probability 8.08% and 46 raw top-decile cells; Washington has mean conservative probability 9.02% and 31 raw top-decile cells. County averages do not imply uniform conditions within a county; the mapped pattern is clustered, and panel scales differ.

2019 | Historical held-out wildfire occurrence

Top: conservative annual recorded-occurrence probability per cell. Bottom: raw ranking percentile, 100 = highest.
Development 2001-2021; held-out maps 2013-2021. Five geographic fits; common legends across years.



Eligible: 931/997. Excluded: 0 unknown labels, 66 flagged zeros. Raw top 10%: 94 cells across all three counties.
Area audit, pooled 2013-2021: full 16 km2 cells = 75.0% of eligible cell-years and 98.0% of top-10% selections.
Cell probabilities depend on clipped area; they are not per-km2 hazard. Original probabilities and ranking inputs are unchanged.
Historical development; county footprint includes non-WUI land. Reporting and climate-vintage limits apply. Separate panel extents; EPSG:26912.

2019 accessible map description

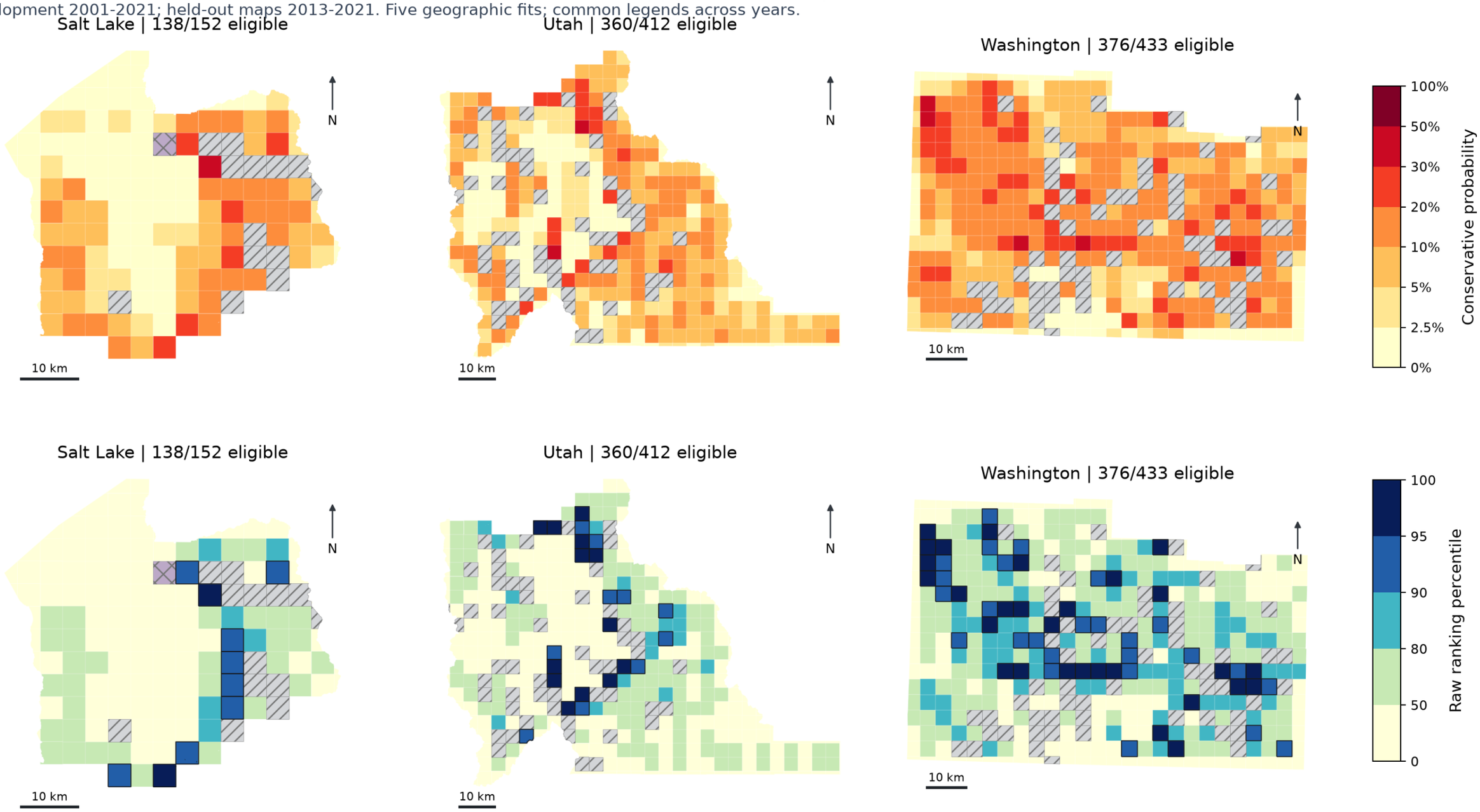
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2019: 931 of 997 original cells have held-out predictions; 0 unknown-label and 66 flagged-zero cells are excluded. Outlines select 94 cells across the three-county cohort. In this year, full cells represent 74.1% of eligible cells and 97.9% of raw top-decile selections. Upper panels show conservative probabilities; lower panels show raw ranking percentiles. Original-area values are retained, and boundary-cell probabilities are not per-km² hazard measures. Development years informed model selection; terrain is a contemporary static proxy.

Geographic comparison among eligible cells: Salt Lake has mean conservative probability 5.80% and 11 raw top-decile cells; Utah has mean conservative probability 6.49% and 26 raw top-decile cells; Washington has mean conservative probability 8.71% and 57 raw top-decile cells. County averages do not imply uniform conditions within a county; the mapped pattern is clustered, and panel scales differ.

2020 | Historical held-out wildfire occurrence

Top: conservative annual recorded-occurrence probability per cell. Bottom: raw ranking percentile, 100 = highest.
Development 2001-2021; held-out maps 2013-2021. Five geographic fits; common legends across years.



Eligible: 874/997. Excluded: 122 unknown labels, 1 flagged zeros. Raw top 10%: 88 cells across all three counties.
Area audit, pooled 2013-2021: full 16 km2 cells = 75.0% of eligible cell-years and 98.0% of top-10% selections.
Cell probabilities depend on clipped area; they are not per-km2 hazard. Original probabilities and ranking inputs are unchanged.
Historical development; county footprint includes non-WUI land. Reporting and climate-vintage limits apply. Separate panel extents; EPSG:26912.

2020 accessible map description

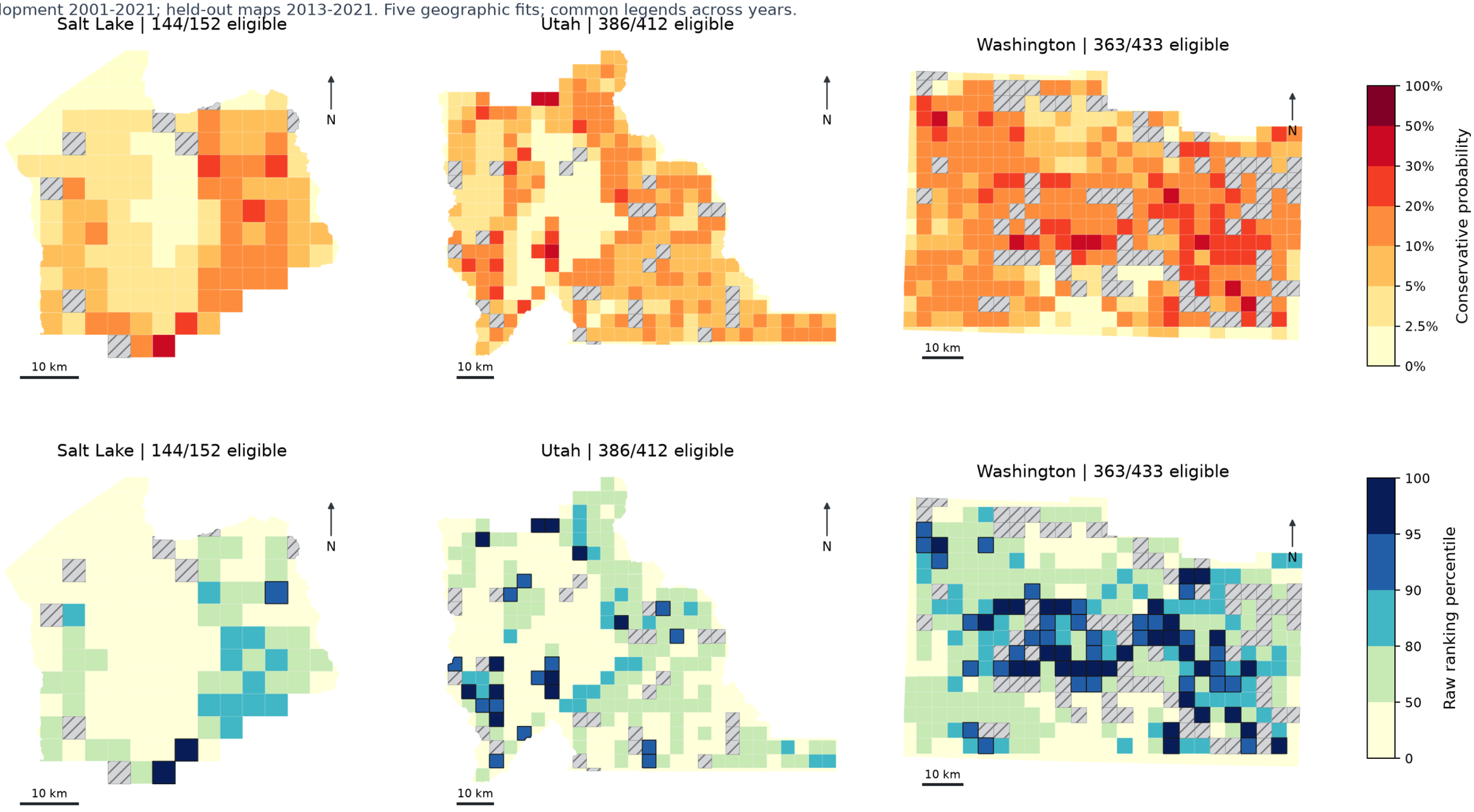
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2020: 874 of 997 original cells have held-out predictions; 122 unknown-label and 1 flagged-zero cells are excluded. Outlines select 88 cells across the three-county cohort. In this year, full cells represent 73.5% of eligible cells and 98.9% of raw top-decile selections. Upper panels show conservative probabilities; lower panels show raw ranking percentiles. Original-area values are retained, and boundary-cell probabilities are not per-km² hazard measures. Development years informed model selection; terrain is a contemporary static proxy.

Geographic comparison among eligible cells: Salt Lake has mean conservative probability 6.17% and 10 raw top-decile cells; Utah has mean conservative probability 6.69% and 23 raw top-decile cells; Washington has mean conservative probability 11.12% and 55 raw top-decile cells. County averages do not imply uniform conditions within a county; the mapped pattern is clustered, and panel scales differ.

2021 | Historical held-out wildfire occurrence

Top: conservative annual recorded-occurrence probability per cell. Bottom: raw ranking percentile, 100 = highest.
Development 2001-2021; held-out maps 2013-2021. Five geographic fits; common legends across years.



Eligible: 893/997. Excluded: 104 unknown labels, 0 flagged zeros. Raw top 10%: 90 cells across all three counties.
Area audit, pooled 2013-2021: full 16 km2 cells = 75.0% of eligible cell-years and 98.0% of top-10% selections.
Cell probabilities depend on clipped area; they are not per-km2 hazard. Original probabilities and ranking inputs are unchanged.
Historical development; county footprint includes non-WUI land. Reporting and climate-vintage limits apply. Separate panel extents; EPSG:26912.

2021 accessible map description

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2021: 893 of 997 original cells have held-out predictions; 104 unknown-label and 0 flagged-zero cells are excluded. Outlines select 90 cells across the three-county cohort. In this year, full cells represent 74.7% of eligible cells and 97.8% of raw top-decile selections. Upper panels show conservative probabilities; lower panels show raw ranking percentiles. Original-area values are retained, and boundary-cell probabilities are not per-km² hazard measures. Development years informed model selection; terrain is a contemporary static proxy.

Geographic comparison among eligible cells: Salt Lake has mean conservative probability 5.93% and 3 raw top-decile cells; Utah has mean conservative probability 7.15% and 24 raw top-decile cells; Washington has mean conservative probability 11.70% and 63 raw top-decile cells. County averages do not imply uniform conditions within a county; the mapped pattern is clustered, and panel scales differ.